This brute-force comparison becomes prohibitively expensive, scaling quadratically, as dataset size grows (for example, comparing 100,000 items requires billions of comparison operations). For large datasets, techniques like locality-sensitive hashing (LSH) are necessary to avoid this computational overload.

Identifying these matches is more than just a data cleaning exercise; it is the key step in constructing a model of reality in your digital twin, from data held in different systems. When your algorithm determines that two different identifiers likely refer to the same real-world object, you are effectively discovering a semantic relationship.

NOTE

Cognite Data Fusion® uses entity matching algorithms based on string similarity matching in its data contextualization tools.

5.2.3 The importance of context in generative AI

Generative AI (GenAI), especially large language models (LLMs), is transformative for digital twins, but its value is realized only with the right context. Without the rich data from a digital twin, an LLM is a general tool, but with context, it becomes a domain-specific expert.

Context is crucial for two reasons:

- *Grounding (preventing hallucination)*: Context anchors the LLM's responses, preventing it from producing inaccurate or fabricated information (hallucination). A general LLM lacks specific knowledge of a plant's layout or assets. If an operator asks, "What's the optimal pressure for Pump A-7?", a general LLM might give a generic or incorrect answer. By feeding the LLM real-time and historical data from the twin (model, age, maintenance records), its response is grounded in reality and relevant to that specific equipment.
- *Specialization (enabling expert analysis)*: Context allows a general-purpose model to specialize. A digital twin contains proprietary domain knowledge (for example, engineering specifications, sensor data, tacit